SOFTWARE PRODUCT · DOCUMENTATION

Athlete Central

Sports & booking

A multi-facility sports booking platform with shared scheduling, booking, pricing, POS, web, mobile, and administration concerns.

ROLE	PERIOD	PLATFORMS
Junior Full-Stack Engineer	**May 2025 – Sept 2025**	**Web · Mobile · POS · Admin**

CONTENTS

1. Overview
2. Scope and personal contribution
3. Product surfaces
4. System architecture
5. Features and status
6. Engineering challenges
7. Technology stack
8. Engineering decisions
9. System scale
10. Evidence, status and limitations

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1. Overview

A multi-facility sports booking platform with shared scheduling, booking, pricing, POS, web, mobile, and administration concerns.

PROJECT	Athlete Central
TYPE	Sports & booking
ROLE	Junior Full-Stack Engineer
PERIOD	May 2025 – Sept 2025
PLATFORMS	Web · Mobile · POS · Admin

2. Scope and personal contribution

> Counts such as "115 Edge Functions" describe the whole platform, not only Prince's own work. His own contributions are the areas of ownership listed in this section.

Documented as built: Designed a shared Firebase Cloud Functions backend across customer web/mobile, Android POS, and Electron admin clients. Built a multi-facility, multi-date booking cart with continuous time-slot merging and availability rules. Centralized standard, peak, weekend, holiday, entrance-fee, and promotional pricing. Separated booking lifecycle from payment lifecycle to keep reservation state explicit and reliable.

3. Product surfaces

Customer Web
`Next.js`

A client of the shared Firebase Cloud Functions backend.

Customer Mobile
`React Native`

A client of the shared Firebase Cloud Functions backend.

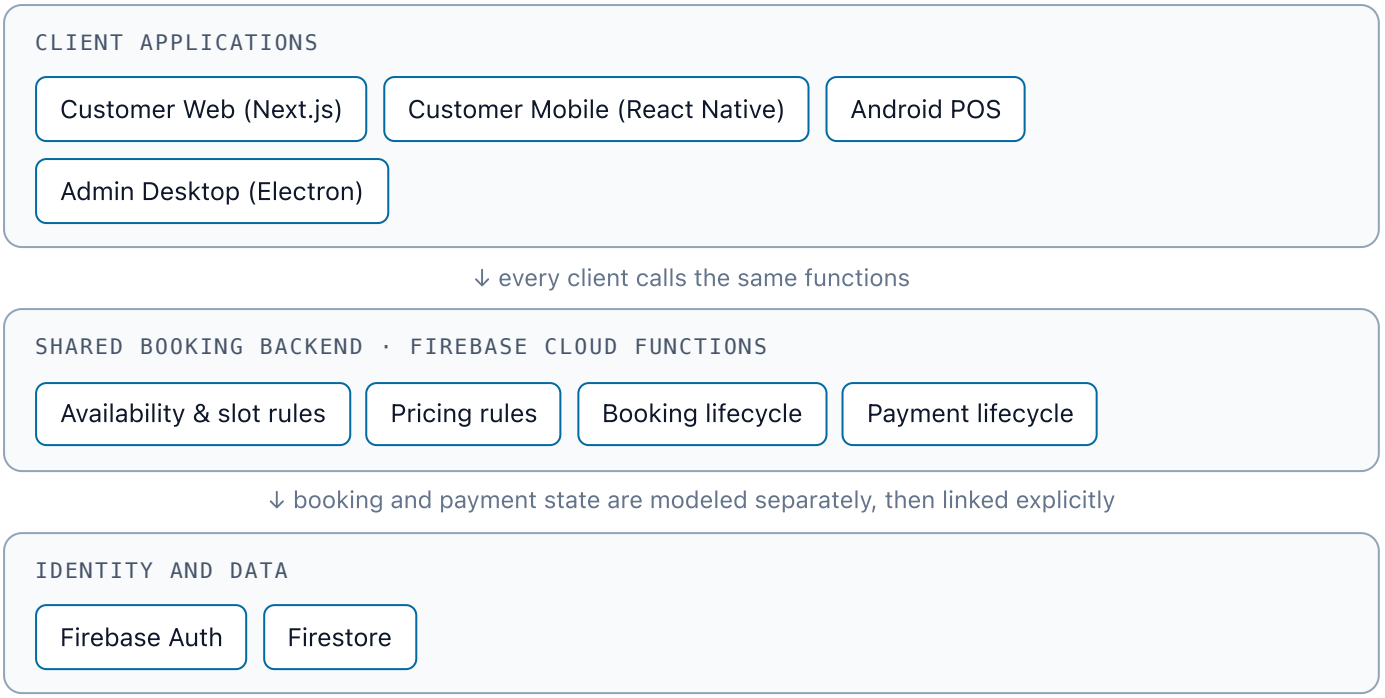
Android POS
`Android`

A client of the shared Firebase Cloud Functions backend.

Admin Desktop
`Electron`

A client of the shared Firebase Cloud Functions backend.

4. System architecture



> Architecture as described in the project record. Per-client feature depth is not itemized.

5. Features and status

Capabilities are labelled by what the project record supports.

Booking

Multi-facility, multi-date cart IMPLEMENTED

A customer assembles reservations across several facilities and dates in a single cart.

Continuous time-slot merging IMPLEMENTED

Adjacent slots are merged into one continuous booking, governed by availability rules.

Live availability IMPLEMENTED

Availability rules are evaluated by the shared backend rather than by each client.

Pricing

Centralized pricing rules IMPLEMENTED

Standard, peak, weekend, holiday, entrance-fee, and promotional pricing live in one place.

Lifecycle & platform

Booking and payment lifecycles kept separate IMPLEMENTED

Reservation state and payment state are modeled independently and connected explicitly.

Shared Firebase Cloud Functions backend IMPLEMENTED

One backend serves customer web and mobile, the Android POS, and the Electron admin client.

6. Engineering challenges

Many carts, many dates, one booking

PROBLEM

Customers reserve several facilities across several dates, and neighbouring slots need to behave as one continuous reservation.

APPROACH

Built a multi-facility, multi-date booking cart that merges continuous time slots and applies availability rules.

IMPACT

Cart behaviour follows availability rules instead of per-screen logic.

Pricing that varies by day and promotion

PROBLEM

Standard, peak, weekend, holiday, entrance-fee, and promotional pricing all apply to the same booking.

APPROACH

Centralized every pricing variant behind one set of backend rules.

IMPACT

Each client reads the same price from the same place.

7. Technology stack

Next.js, React Native, Electron, Firebase, TypeScript, Tailwind CSS.

8. Engineering decisions

Shared booking domain

PROBLEM

Multiple client applications need identical scheduling and pricing behavior.

DECISION

Centralize booking rules and domain models behind shared cloud functions.

IMPACT

Web, mobile, POS, and admin clients consume one consistent scheduling source of truth.

Booking/payment separation

PROBLEM	DECISION	IMPACT
Payment state and reservation state change independently.	Model booking and payment lifecycles separately and connect them explicitly.	Failed or delayed payments do not corrupt the operational state of reservations.

9. System scale

- Multi-facility booking
- Multi-date cart
- Live availability
- Dynamic pricing
- Shared backend

> These figures describe the whole system. His own contributions are the areas of ownership listed in this section.

10. Evidence, status and limitations

- This document describes architecture and behaviour only. It contains no credentials, internal endpoints or customer data.
- No performance, revenue or user-count figures are published for this project.
- Only the capabilities labelled Implemented below are documented. No in-progress or planned items are documented for this project.
- Product screenshots are not published yet; the case study shows a project poster only.

Sources:

- Case study: https://pjdevph.vercel.app/projects/athlete-central